

Variables & Data Types

Giving Values a Label That Can Change

THE BIG QUESTION

If a program needs to remember something that changes — a score, an age, an answer — how does it hold onto it?

Reopening Two Boxes

CLASS 05 LEFT OPEN

Fixed Text

Every program said the exact same thing, every single time — changing it meant editing the code and running it again.

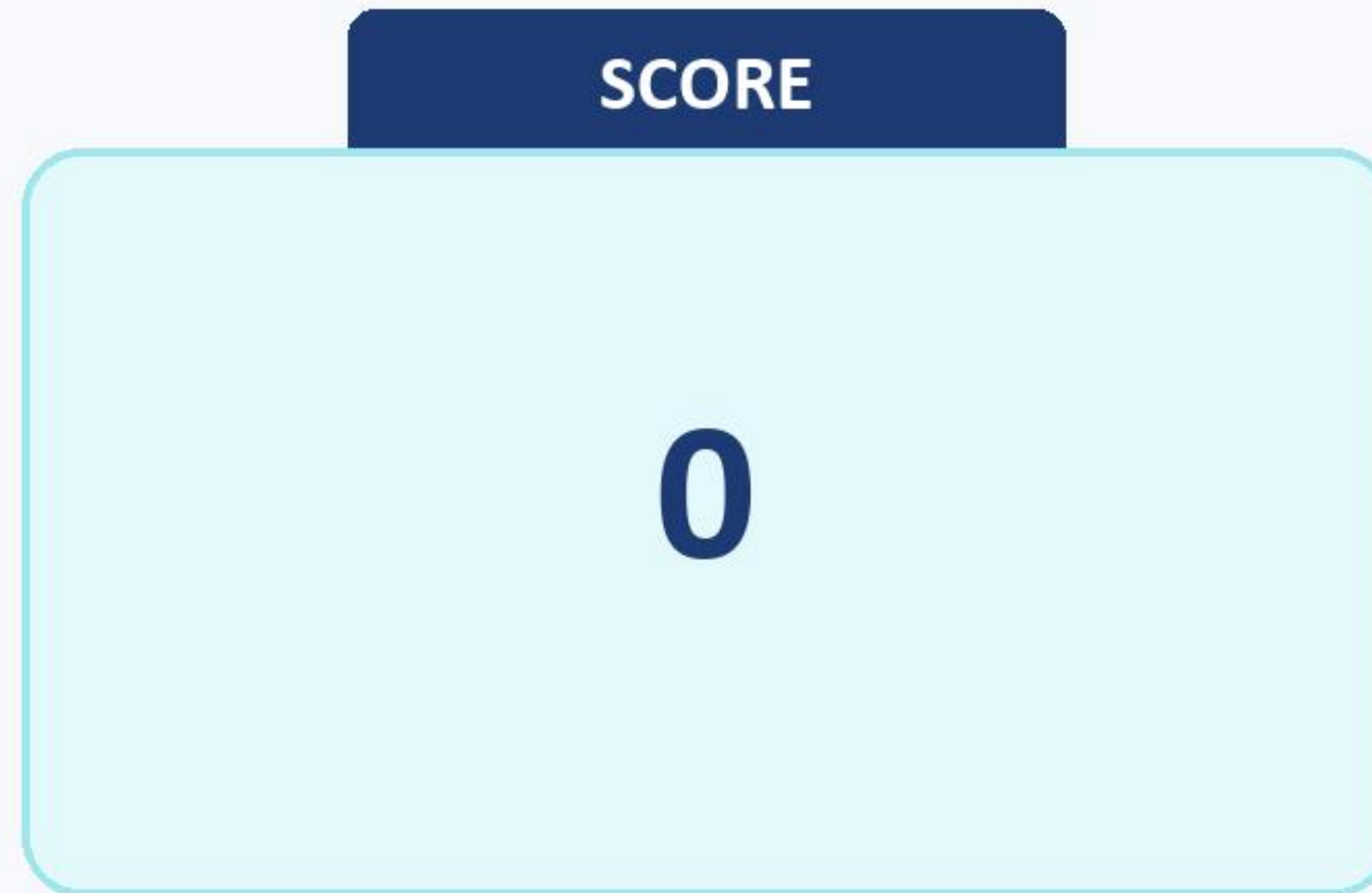
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CLASS 04 REOPENED

Bits

Information is represented inside a computer as bits — you never see them directly.

What Is a Variable?



score = 0 → print(score) → shows 0

= Means Store, Not Equals

score = 0

READ AS: "score GETS 0"

NOT: "score equals 0"

One Name, Two Different Values

- ```
score = 0
print(score)
score = 10
print(score)
```

OUTPUT

0

10

# print(name) vs. print("name")



```
name = "Priya"
```

```
print(name)
```

NO QUOTES → THE STORED VALUE

Priya



```
name = "Priya"
```

```
print("name")
```

QUOTES → THE LITERAL WORD

name

# Reassignment

- ```
score = 0
print(score)
score = 10
print(score)
score = 25
print(score)
```

OUTPUT — the old value is never seen again

```
0
10
25
```

Naming Variables

VALID

- `age = 18`

`student_name = "Rio"`

`_temp = 5`

INVALID

- `2nd_place = ...`

`student name = ...`

`class = ...`

Starts with a digit · contains a space · a reserved Python word — three different problems, all invalid.

Meet the Data Types

Variable	Value	Type
age	18	int
price	19.99	float
name	"Priya"	str
is_student	True	bool

```
print(type(age)) → <class 'int'>
```

Numbers: int vs. float

INT — A WHOLE NUMBER



apples = 3

FLOAT — HAS A DECIMAL POINT



temperature = 36.6

Strings, Revisited

- ```
city = "Kolkata"

print(city)
```

A VARIABLE — NO QUOTES

Kolkata

- ```
city = "Kolkata"  
  
print("city")
```

LITERAL TEXT — QUOTES

city

Booleans

- ```
is_raining = True
is_weekend = False
print(is_raining)
```

OUTPUT

True

# Combining Text and Variables



```
name = "Priya"
```

```
age = 18
```

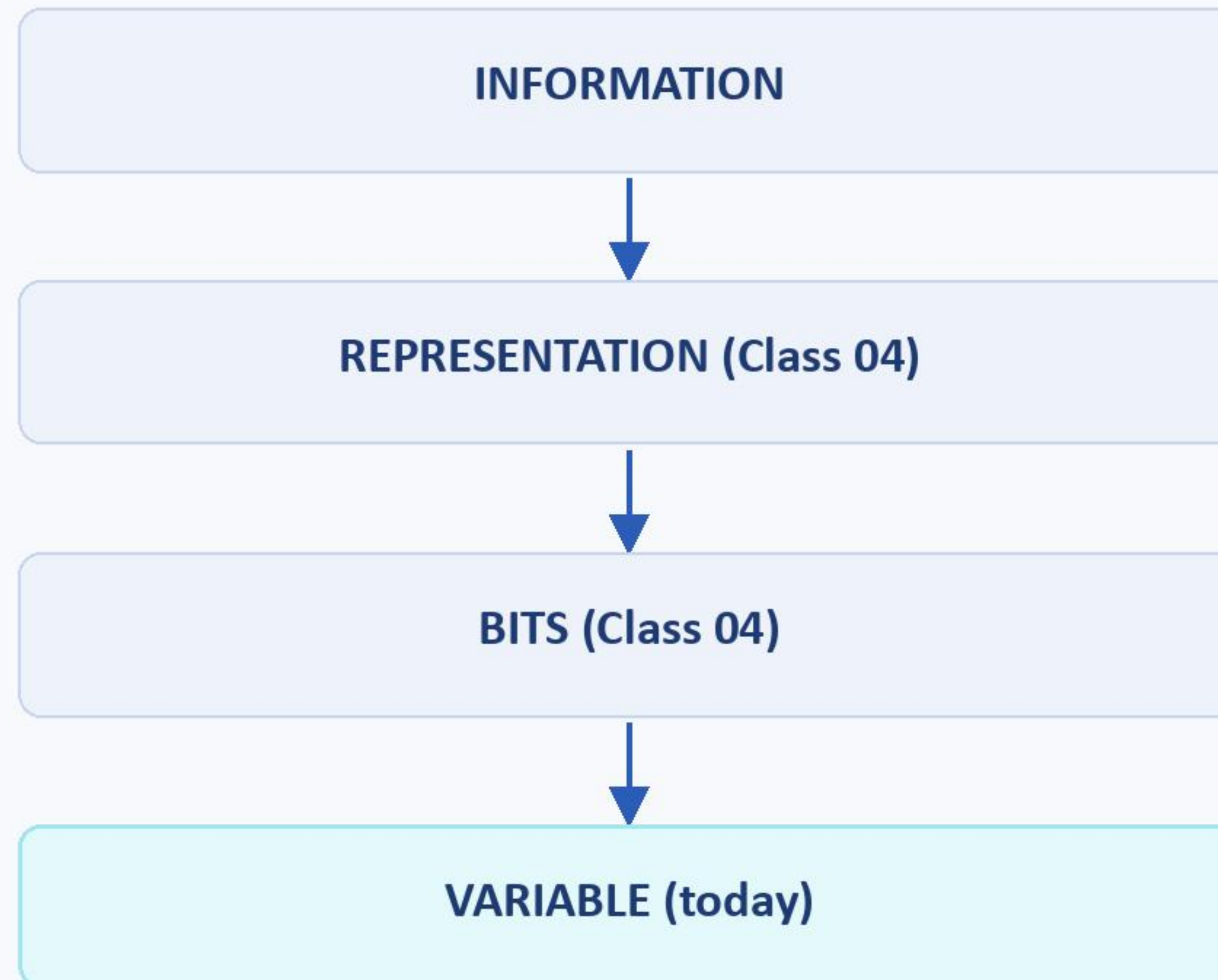
```
print("My name is", name, "and I am", age, "years old.")
```

## OUTPUT

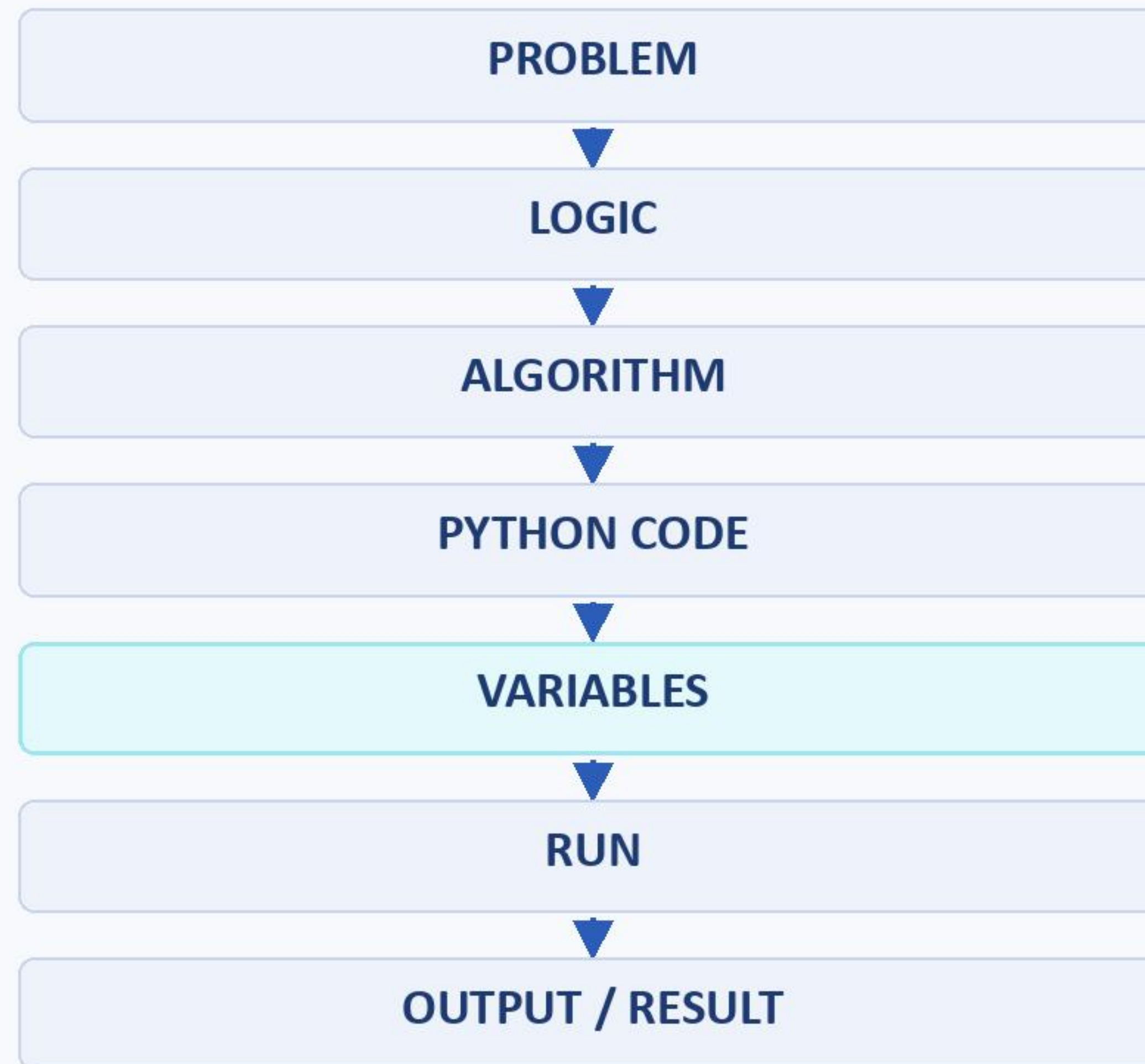
My name is Priya and I am 18 years old.

# A Variable Is a Labeled Representation

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## THE HERO DIAGRAM



*Cyan = new today · Card = already known from Classes 01 and 05*

## Variable Profile Card

At least four variables:

one string, one int, one float, one boolean.

*Then print() each one in a full sentence, combining text and variables with commas.*

# Variable Profile Card — Template

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```
name = "____"
```

```
age = ____
```

```
height = ____
```

```
is_student = ____
```

```
print("My name is", name)
```

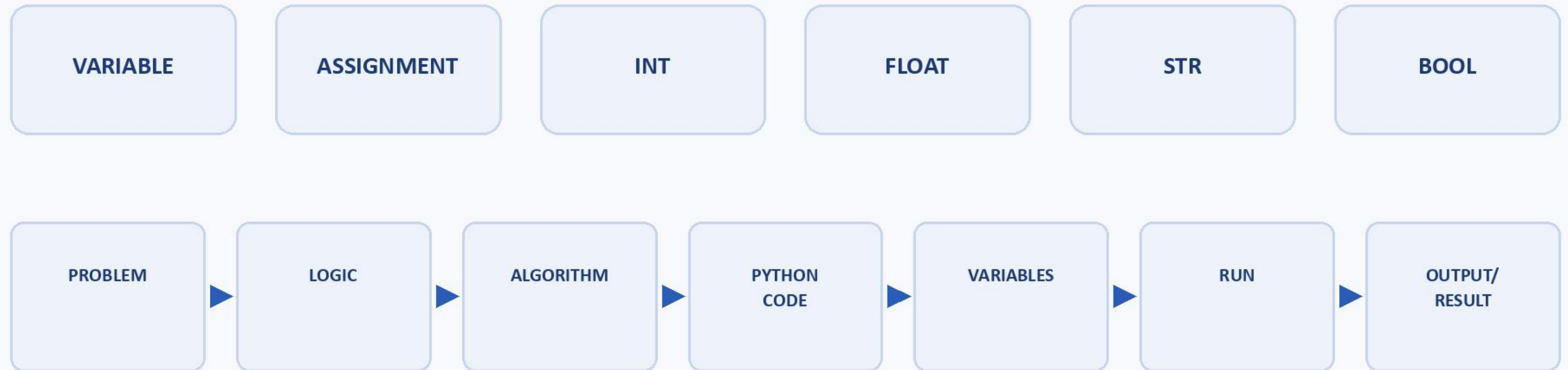
## REFLECT

**The instant you reassign a variable, what happens to its old value?**

*Completely replaced. Nothing is kept.*

# Quick Recap

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## A QUESTION FOR NEXT TIME

**We can print numbers now — but we haven't done any math with them. What happens when a program needs to add, compare, or calculate?**

#### FINAL TAKEAWAY

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**“Last class, your programs could only ever say the exact same thing. Today, they can hold something, change it, and tell you about it. That one idea is what makes a program feel alive instead of frozen.”**